

2.3 Conditioning on an arbitrary random variable

Definition. Let $\xi \in L^1$, and let η be any random variable. A conditional expectation of ξ given η is an integrable random variable, denoted by $E(\xi|\eta)$, with the following properties:

1. $E(\xi|\eta)$ is $\sigma(\eta)$ -measurable.
2. For every $A \in \sigma(\eta)$, it satisfies

$$\int_A E(\xi|\eta) dP = \int_A \xi dP.$$

Conditional Probability. For $A \in \mathcal{F}$, the conditional probability of A given η is defined by

$$P(A|\eta) = E(\mathbb{1}_A|\eta).$$

Lemma 2.1. Let $\mathcal{G} \subset \mathcal{F}$ be a σ -field, and let ξ be an integrable, and $\mathcal{G}$ -measurable random variable. If

$$\int_B \xi dP = 0 \quad \text{for all } B \in \mathcal{G},$$

then

$$\xi = 0 \quad \text{a.s.} \iff P(\xi \neq 0) = 0.$$

Proof. Note that

$$\{\xi \neq 0\} = \bigcup_{n \geq 1} \{|\xi| > \frac{1}{n}\} \cup \bigcup_{n \geq 1} \{|\xi| < -\frac{1}{n}\}.$$

For $\epsilon > 0$, the set $B_+ = \{\xi \geq \epsilon\}$ belongs to $\mathcal{G}$. By assumption,

$$0 = \int_{B_+} \xi dP \geq \int_{B_+} \epsilon dP = \epsilon P(B_+).$$

So, we have $P\{\xi \geq \epsilon\} = 0$. Taking $\epsilon = \frac{1}{n}$, we obtain

$$P\{\xi \neq 0\} \leq \sum_{n \geq 1} (P\{|\xi| > \frac{1}{n}\} + P\{\xi < -\frac{1}{n}\}) = 0.$$

□

Remark. Any two random variable satisfying the definition of conditional expectation are equal a.s. In particular,

$$\xi = \xi' \quad \text{a.s.} \implies E(\xi|\eta) = E(\xi'|\eta) \quad \text{a.s.}$$

Proof. Let Z, Z' satisfy the definition of conditional expectation.¹ Then, Lemma 2.1 gives $Z = Z'$ a.s. If $\xi = \xi'$ a.s. their integral over A agree, so the same argument gives

$$E(\xi|\eta) = E(\xi'|\eta) \quad \text{a.s.}$$

□

¹Note that their difference is integrable and $\sigma(\eta)$ -measurable by Proposition 1. Moreover, for all $A \in \sigma(\eta)$, we have

$$\int_A (Z - Z') dP = \int_A Z dP - \int_A Z' dP = \int_A \xi dP - \int_A \xi dP = 0.$$

Exercise 2.4. Take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, and P be the Lebesgue measure $m(\cdot)$. Define $\xi(x) = 2x^2$ and

$$\eta(x) = \begin{cases} 2, & 0 \leq x < \frac{1}{2} \\ x, & \frac{1}{2} \leq x \leq 1 \end{cases}$$

Find $E(\xi|\eta)$.

Proof. Since it is $\sigma(\eta)$ -measurable, Doob-Dynkin lemma shows that

$$E(\xi|\eta) = h(\eta) \quad \text{for some Borel function } h.$$

Since $\{\eta = 2\} = [0, \frac{1}{2})$, condition 2 gives

$$\int_{\{\eta=2\}} h(\eta) dP = \int_{\{\eta=2\}} \xi dP$$

where

$$\int_{\{\eta=2\}} h(\eta) dP = \int_0^{\frac{1}{2}} h(2) dP = \frac{1}{2} h(2) \quad \text{and} \quad \int_{\{\eta=2\}} \xi dP = \int_0^{\frac{1}{2}} 2x^2 dP = \frac{1}{12}.$$

Hence, $h(2) = \frac{1}{6}$. Next, let $B \in \mathcal{B}[\frac{1}{2}, 1]$, then $\{\eta \in B\} = B$ and condition 2 gives

$$\left(\int_B h(\eta) dP = \right) \int_B h(x) dP = \int_B 2x^2 dP (= \int_B \xi dP).$$

Since B is arbitrary, we have $h(x) = 2x^2$ a.e. in $[\frac{1}{2}, 1]$. Define

$$h(x) = \begin{cases} \frac{1}{6}, & x = 2 \\ 2x^2, & \frac{1}{2} \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad Z = \begin{cases} h(\eta) = \frac{1}{6}, & 0 \leq x < \frac{1}{2} \\ h(\eta) = 2x^2, & \frac{1}{2} \leq x \leq 1 \end{cases}$$

We now verify that $Z = E(\xi|\eta)$ a.s. :

1. Since h is Borel, $Z = h(\eta)$ is $\sigma(\eta)$ -measurable.
2. Moreover, Z is bounded, so it is integrable.
3. We have already seen that

$$\int_{[0, \frac{1}{2})} Z dP = \int_{[0, \frac{1}{2})} \xi dP \quad \text{and} \quad \int_B Z dP = \int_B \xi dP \quad \text{for all } B \in \mathcal{B}[\frac{1}{2}, 1].$$

4. Every event in $\sigma(\eta)$ is either B or $[0, \frac{1}{2}) \cup B$.

Hence, by additivity of the integral, we have

$$\int_A Z dP = \int_A \xi dP \quad \text{for all } A \in \sigma(\eta).$$

□

Exercise 2.6. On $[0, 1]$ with the Lebesgue measure, let $\xi(x) = 2x^2$ and $\eta(x) = 1 - |2x - 1|$. Find $E(\xi|\eta)$.

Exercise 2.7. Let $f_{\xi, \eta}(x, y) = x + y$ on $[0, 1]^2$. Find $E(\xi|\eta)$.

Proof. We write $Z = h(\eta)$ for some Borel function h . For any $B \in \mathbb{B}$,

$$\int_{\{\eta \in B\}} Z dP = \int_{\{\eta \in B\}} \xi dP.$$

Indeed,

$$\begin{aligned} \int_{\{\eta \in B\}} Z dP &= \int_B \int_{\mathbb{R}} h(y) f_{\xi, \eta}(x, y) dx dy \\ &= \int_B h(y) \int_0^1 (x + y) dx dy \\ &= \int_B h(y) \left(\frac{1}{2} + y \right) dy \end{aligned}$$

and

$$\begin{aligned} \int_{\{\eta \in B\}} \xi dP &= \int_B \int_{\mathbb{R}} x f_{\xi, \eta}(x, y) dx dy \\ &= \int_B \int_0^1 x(x + y) dx dy \\ &= \int_B \left(\frac{1}{3} + \frac{1}{2}y \right) dy. \end{aligned}$$

Hence, we have

$$h(y) = \frac{\frac{1}{3} + \frac{1}{2}y}{\frac{1}{2} + y} = \frac{2 + 3y}{3(1 + 2y)} \quad \text{a.e. on } [0, 1].$$

We redefine

$$h(y) = \begin{cases} \frac{2+3y}{3(1+2y)}, & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

□